

HS 651: Advanced Econometrics

Course Code	HS 651
Course Name	Advanced Econometrics
L-T-P-C	3-0-0-3
Prerequisites	Basic Econometrics
Intended for	Ph. D.
Approval	6th Senate (May 2014)

Course Contents

- **Regression analysis** Historic origin and modern interpretation of regression, simple, two variable regression models, the method of ordinary least square, the Gauss-Markov theorem, the coefficient of determination. [8 Lectures]
- **Multiple regression analysis** three variable model, OLS and ML estimation of the partial regression coefficients, the Cobb-Douglas production function, polynomial regression models, dummy variable regression models; ANOVA and ANCOVA models, the linear probability model, the logit and probit models, the Poisson and negative binomial regression models, maximum likelihood estimation, the generalized method of moments. [12 Lectures]
- **Panel data regression model** pooled OLS regression, the fixed effects least squares dummy variable (LSDV) model, the fixed effect within group (WG) estimator, the random effects model (REM), different-in-different approach, regression discontinuity, propensity score matching. [10 Lectures]
- **Time series econometrics** stochastic processes, spurious regression, test of stationarity, unit root test, transforming non stationary time series, economic forecasting; AR, MR and ARIMA modeling of time series data, Box- Jenkins methodology, vector auto regression (VAR), testing causality using VAR: the Granger causality test, impulse response function, measuring volatility in financial time series; the ARCH and GARCH models. [12 Lectures]

Course Readings

1. Jeffrey M. Wooldridge, **Introductory Econometrics: A Modern Approach**, 4th Edition (Paperback), Cengage Learning India, 2012.
2. William H. Greene, **Econometric Analysis**, 5th Edition (Paperback), Pearson Education, 2003.
3. Jeffrey M. Wooldridge, **Econometric Analysis of Cross Section and Panel Data**, 2nd Edition, MIT Press, 2010.
4. Dougherty, Christopher, **Introduction to Econometrics**, 4th Edition, 2011.
5. James D. Hamilton, **Time Series Analysis**, Princeton University Press, 1994.
6. Maddala, G. S., **Econometrics**, Mc Graw Hill, 1997.

Readings

1. Gujarati, D. N., **Basic Econometrics**, 5th Edition, Mc Graw Hill, 2012.
2. Intriligator M. D., **Econometric Methods, Techniques and Applications**, Prentice Hall, 1997.
3. Johnston, J., **Econometric methods**, Mcgraw Hill book Co., 1991.
4. Franses P. H., **Time Series Models for Business and Economic forecasting**, Cambridge Press, 1998.
5. Krishna, K. L., **Econometric Applications in India**, Oxford University Press, 1997.
6. Kennedy, P., **A Guide to Econometrics**, 4th Edition, MIT press, 1998.
7. Goldberger, A. S., **Introductory Econometrics**, Harvard University press, 1998.